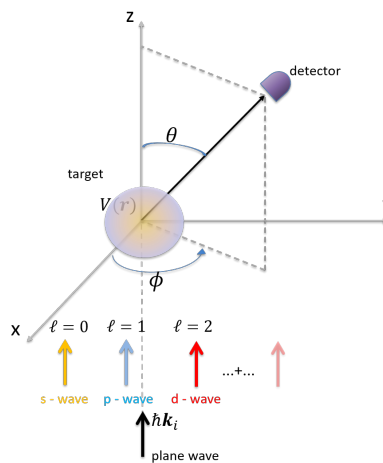


LECTURE 9: Partial waves, phase shifts and the optical theorem

We specialize to spherically symmetric potentials $V(\mathbf{r}) = V(r)$ where the scattering amplitude has only a θ angle dependence. We decompose the incoming wave into independent angular-momentum channels (partial waves). This is illustrated in the figure below, showing that a plane wave can be decomposed into a sum over an infinite number of spherical harmonics with index $\ell = 0$ (s-wave), $\ell = 1$ (p-wave) and so on. We will make the notation k instead of k_i for both the incoming and outgoing momenta (elastic scattering), to simplify notation.



Main takeaways

- (i) It is convenient to decompose the plane wave into spherical components and treat the scattering in each partial wave independently.
- (ii) Each partial wave acquires a phase shift $\delta_\ell(k)$, and these phase shifts fully determine the scattering amplitude $f(\theta)$.
- (iii) Low energy $\Rightarrow$ large $\lambda = 2\pi/k_i$: the particle cannot resolve structure; only the s-wave ($\ell = 0$) survives. Scattering = pure diffraction off the whole shape.

$$f(\theta) = \frac{1}{k} e^{i\delta_0(k)} \sin \delta_0(k),$$

and $\sigma_{\text{tot}} = 4\pi a_s^2$, with a_s being the scattering length.

- (iv) Higher energy $\Rightarrow$ smaller λ : the particle can resolve finer angular structure. Many partial waves contribute and the total cross section approaches the geometric value $\sigma_{\text{tot}} = 4\pi R^2$, where R is the physical size of the scatterer.

Before we tackle the maths, let us first highlight some of the important steps and key points.

Partial-wave expansion

The **key point** is that we can decompose any plane wave exactly into a superposition of spherical harmonics:

$$e^{ikz} = \sum_{\ell=0}^{\infty} i^{\ell} (2\ell + 1) j_{\ell}(kr) P_{\ell}(\cos \theta).$$

To make it clear, for first two orders we have

$$e^{ikz} = \underbrace{\frac{\sin(kr)}{kr}}_{\ell=0 \text{ term}} + \underbrace{3i \left[\left(\frac{\sin(kr)}{(kr)^2} - \frac{\cos(kr)}{kr} \right) \cos \theta \right]}_{\ell=1 \text{ term}} + \underbrace{-\frac{5}{2} \left[\frac{3 \sin(kr)}{(kr)^3} - \frac{\sin(kr)}{kr} - \frac{3 \cos(kr)}{(kr)^2} \right] (3 \cos^2 \theta - 1)}_{\ell=2 \text{ term}} + \dots$$

Shifts in the absence of a target (no scattering)

No target - Notice the following aspects (in the absence of a target $V(r) = 0$):

- only the terms falling off slowly with $1/r$ survive in the far field limit
- **even** orders have zero shift while **odd** orders have $\pi/2$ shifts. This can be seen generally formulated in terms of the far field expansion of the Bessel functions

$$j_{\ell}(kr) \longrightarrow \frac{1}{kr} \sin\left(kr - \frac{\ell\pi}{2}\right),$$

Shifts in the presence of a target potential

With target - Adding a target $V(r)$, what will happen is:

- Each ℓ -component scatters independently - so we can think independently of each ℓ order
- Each free spherical Bessel function is shifted

$$j_{\ell}(kr) \longrightarrow \frac{1}{kr} \sin\left(kr - \frac{\ell\pi}{2} + \delta_{\ell}(k)\right),$$

which defines the holy grail of this lecture, the phase shift $\delta_{\ell}(k)$.

- We can build the full asymptotic scattering state wavefunction:

$$\psi^{(+)}(r, \theta) = \sum_{\ell=0}^{\infty} (2\ell + 1) P_{\ell}(\cos \theta) \frac{1}{2ikr} \left(e^{-ikr} - e^{2i\delta_{\ell}(k)} e^{ikr} \right).$$

- This allows to compute the scattering amplitude and subsequently the cross section by knowing only the phase shifts. However, the problem now is to see how one can compute this, which we will describe in the following

Partial waves and phase shifts

We consider elastic scattering of a particle of momentum k off a spherically symmetric potential $V(r)$. The stationary Schrödinger equation which we can write for the asymptotic solution wavefunction is

$$\left[-\frac{\hbar^2}{2m} \nabla^2 + V(r) \right] \psi(\mathbf{r}) = \frac{\hbar^2 k^2}{2m} \psi(\mathbf{r}).$$

From the previous lecture, we know that the far field solution (the asymptotic solution) with final k equal to the incoming momentum is

$$\psi^{(+)}(r, \theta) \xrightarrow{r \rightarrow \infty} e^{ikz} + f(\theta) \frac{e^{ikr}}{r}.$$

We also know that the initial state can be expressed as

$$e^{ikz} = \sum_{\ell} (2\ell+1) i^{\ell} j_{\ell}(kr) P_{\ell}(\cos \theta),$$

so we will try to figure out how to write the asymptotic solution in a similar fashion

$$\psi^{(+)}(r, \theta) = \sum_{\ell=0}^{\infty} (2\ell+1) P_{\ell}(\cos \theta) \frac{u_{\ell}(r)}{kr}.$$

The task now is to determine the radial functions $u_{\ell}(r)$.

Radial equation

Replacing the solution $\psi^{(+)}(r, \theta)$ in this form into the Schrödinger equation yields, for each ℓ ,

$$\frac{d^2 u_{\ell}}{dr^2} + \left[k^2 - \frac{\ell(\ell+1)}{r^2} - \frac{2m}{\hbar^2} V(r) \right] u_{\ell}(r) = 0.$$

We will sketch here the steps in the procedure for finding these $u_{\ell}(r)$. We will assume an **inside** region of finite interaction potential is separated at a boundary a to the **outside** which the potential is zero. Matching the two solutions at the boundary will provide a way to find the phase shifts.

(i) **Inside solution $u_{\ell}^{(in)}(r)$ (potential region).**

Solve for $0 < r < a$:

$$\frac{d^2 u_{\ell}^{(in)}(r)}{dr^2} + \left[k^2 - \frac{\ell(\ell+1)}{r^2} - \frac{2m}{\hbar^2} V(r) \right] u_{\ell}^{(in)}(r) = 0.$$

We need to impose regularity at the origin:

$$u_{\ell}^{(in)}(r) \propto r^{\ell+1} \quad (r \rightarrow 0),$$

which is necessary such that the solution does not diverge at the origin.

(ii) **Outside solution $u_{\ell}^{(out)}(r)$ (free region).**

For $r \geq a$, where $V = 0$, the solution can be written in terms of **spherical Bessel j_{ℓ} and Neumann n_{ℓ} functions**

$$u_{\ell}^{(out)}(r) = A_{\ell} [\cos \delta_{\ell} j_{\ell}(kr) - \sin \delta_{\ell} n_{\ell}(kr)],$$

which automatically has the correct asymptotic form $\sin(kr - \ell\pi/2 + \delta_{\ell})$. We define the phase shift $\delta_{\ell}(k)$ through the asymptotic form

$$u_{\ell}^{(out)}(r) \xrightarrow{r \rightarrow \infty} \sin\left(kr - \frac{\ell\pi}{2} + \delta_{\ell}(k)\right).$$

(iii) **Logarithmic derivative at the boundary**

At a radius a where $V(a) \approx 0$, compute

$$L_{\ell}(k) = \frac{u_{\ell}'^{(in)}(r)}{u_{\ell}^{(in)}(r)} \Big|_{r=a} = \frac{u_{\ell}'^{(out)}(r)}{u_{\ell}^{(out)}(r)} \Big|_{r=a}.$$

(iv) **Matching conditions and phase shift.**

Matching logarithmic derivatives gives

$$L_\ell(k) = k \frac{j'_\ell(ka) - \tan \delta_\ell(k) n'_\ell(ka)}{j_\ell(ka) - \tan \delta_\ell(k) n_\ell(ka)}.$$

(v) Solving for the phase shift yields the master formula:

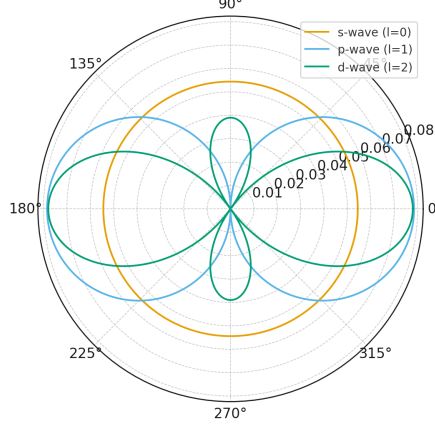
$$\tan \delta_\ell(k) = \frac{k j'_\ell(ka) - L_\ell(k) j_\ell(ka)}{k n'_\ell(ka) - L_\ell(k) n_\ell(ka)}.$$

This is a formula where the shifts are determined assuming we can somehow find this function $L_\ell(k)$ - which typically gets relegated to a numerical problem.

(vi) Far outside the potential ($r \gg a$),

$$\psi_\ell^{(+)}(r, \theta) \xrightarrow{r \rightarrow \infty} (2\ell + 1) P_\ell(\cos \theta) \frac{1}{kr} \sin\left(kr - \frac{\ell\pi}{2} + \delta_\ell(k)\right).$$

Angular patterns of s, p, d partial waves at $kr=10$



Scattering amplitude in terms of phase shifts. The optical theorem.

Using this asymptotic form, it is easy to identify the scattering amplitude as

$$f(\theta) = \frac{1}{k} \sum_{\ell=0}^{\infty} (2\ell + 1) e^{i\delta_\ell(k)} \sin \delta_\ell(k) P_\ell(\cos \theta).$$

As you see in figure above, the different orders have very different patterns.

Integrating over all solid angles and using the orthogonality of the Legendre polynomials

$$\int d\Omega P_\ell(\cos \theta) P_{\ell'}(\cos \theta) = \frac{4\pi}{2\ell + 1} \delta_{\ell\ell'}.$$

yields

$$\sigma_{\text{tot}} = \int d\Omega |f(\theta)|^2 = \frac{4\pi}{k^2} \sum_{\ell=0}^{\infty} (2\ell + 1) \sin^2 \delta_\ell(k).$$

Notice that the fact that the total cross section is an area is obvious from using $\lambda_{\text{dB}} = h/(\hbar k) = 2\pi/k$.

The other cool thing is that the same quantity can be obtained from evaluating $f(\theta)$ at $\theta = 0$ (using $P_\ell(1) = 1$) and taking the imaginary part:

$$f(0) = \frac{1}{k} \sum_{\ell=0}^{\infty} (2\ell+1) e^{i\delta_\ell} \sin \delta_\ell \quad \rightarrow \quad \Im f(0) = \frac{1}{k} \sum_{\ell=0}^{\infty} (2\ell+1) \sin^2 \delta_\ell(k).$$

This is no accident, but it is known as the **optical theorem** stating that

$$\sigma_{\text{tot}} = \frac{4\pi}{k} \Im f(0) = 2\lambda_{\text{dB}} \Im f(0).$$

How to use it:

- In experiment: measuring (or inferring) the forward scattering amplitude $f(0)$ gives direct access to the *total* cross section, without resolving the full angular distribution.
- In partial-wave language: the optical theorem encodes the statement that flux removed from the incoming beam is exactly accounted for by the sum over all scattered channels (conservation of probability).

A few considerations

Regarding low versus high energy scattering. Let R denote the characteristic range of the potential $V(r)$. A useful dimensionless parameter is

$$kR = 2\pi \frac{R}{\lambda_{\text{dB}}}$$

Low-energy regime: $kR \ll 1$ (s-wave scattering)

In this regime, the de Broglie wavelength is long compared to the range of the potential and one can show that

$$\delta_\ell(k) \propto (kR)^{2\ell+1} \quad \Rightarrow \quad \text{for } kR \ll 1, \text{ s-wave } (\ell = 0) \text{ dominates.}$$

For $\ell = 0$, the scattering amplitude becomes nearly isotropic:

$$f(\theta) \approx f_0 = \frac{1}{k} e^{i\delta_0} \sin \delta_0.$$

At very low energies, it is convenient to introduce the *s-wave scattering length* a_s via the effective-range expansion

$$k \cot \delta_0(k) \approx -\frac{1}{a_s} \quad \Rightarrow \quad f_0(k) \approx -\frac{a_s}{1 + ika_s}.$$

In the strict $k \rightarrow 0$ limit,

$$f_0 \rightarrow -a_s, \quad \sigma_{\text{tot}} \approx 4\pi a_s^2.$$

High-energy regime: $kR \gg 1$

For higher energies, many partial waves contribute significantly. Roughly, partial waves with $\ell \lesssim \ell_{\text{max}} \sim kR$ are important. The number of contributing channels scales with kR , and the angular pattern of the scattering becomes more structured, with oscillations reminiscent of diffraction and classical deflection.

In this regime, the total cross section often approaches a classical geometric scale such as

$$\sigma_{\text{tot}} \sim \pi R^2.$$

Let us now comment on the compatibility of the first Born approximation and s-wave scattering. Even though:

s-wave approximation = low-energy expansion

Born approximation = weak-potential expansion

they become compatible in the domain where both hold simultaneously:

$$V(r) \ll E, \quad kR \ll 1.$$

In that overlapping regime:

$$f(\theta) \approx \frac{1}{k} \sin \delta_0^{(1)}, \quad \delta_0^{(1)} = -\frac{m}{\hbar^2 k} \int_0^\infty dr \, r V(r).$$

So: Born does *not* imply s-wave, and s-wave does *not* imply Born — but the two approximations match perfectly when energy is low and the potential is weak.

Appendix: Scattering off a square-well potential

For a central potential, the scattering amplitude admits the partial-wave representation

$$f(\theta) = \frac{1}{k} \sum_{\ell=0}^{\infty} (2\ell+1) e^{i\delta_{\ell}(k)} \sin \delta_{\ell}(k) P_{\ell}(\cos \theta),$$

At low energies, for a short-range potential such as the square well, the dominant contribution comes from the s-wave ($\ell = 0$):

$$f(\theta) \simeq f_0(k) = \frac{1}{k} e^{i\delta_0(k)} \sin \delta_0(k) \longrightarrow \Im f_0(k) = \frac{1}{k} \sin^2 \delta_0(k).$$

In this regime the scattering is isotropic: the differential cross section is independent of θ . In the weak-potential / low-energy limit ($|V_0| \ll E$, $ka \ll 1$), the phase shift $\delta_0(k)$ is small and we may expand $\sin \delta_0 \simeq \delta_0$. One finds from the exact expression

$$\delta_0(k) = -ka + \arctan \left[\frac{k}{k_1} \tan(k_1 a) \right]$$

that, to first order in V_0 ,

$$\delta_0(k) \simeq -\frac{2mV_0}{3\hbar^2} a^3 k.$$

Thus

$$\sigma_{\text{tot}} \simeq \frac{4\pi}{k} \Im f_0(k) = \frac{4\pi}{k^2} \sin^2 \delta_0(k) = \frac{16\pi}{9} \frac{m^2 V_0^2 a^6}{\hbar^4}.$$

Connection to the previously obtained result - We can now connect this Born result to the partial-wave / phase-shift description. Recall the first Born amplitude for the square-well potential:

$$f^{(1)}(\theta) = -\frac{2m}{\hbar^2} V_0 a^3 \frac{1}{(qa)^2} \left(\frac{\sin(qa)}{qa} - \cos(qa) \right), \quad q = 2k \sin \frac{\theta}{2}.$$

For $ka \ll 1$ we also have $qa \ll 1$ for all scattering angles. To leading order in qa ,

$$f^{(1)}(\theta) \simeq -\frac{2m}{\hbar^2} V_0 a^3 \frac{1}{3} = -\frac{2mV_0 a^3}{3\hbar^2}.$$

This is *independent of θ* : the Born amplitude becomes isotropic in the low-energy limit.